Optimal Allocation for IMC Prosperity (Research / Scale / Speed)

1 Problem Setup

We allocate proportions of total budget across three components:

$$r + s + p \leq 100, \quad r, s, p \geq 0$$

where:

- r = Research (%)
- s = Scale (%)
- p = Speed (%)

Total budget:

$$\text{Budget} = 50{,}000$$

Objective:

$$\max \ \text{PnL} = R(r) \cdot S(s) \cdot V(p) - 500(r + s + p)$$

2 Model Components

2.1 Research Function

$$R(x) = -108320.888 \cdot \log_{0.0821202}(x + 0.999975) + 0.22793$$

This behaves approximately like:

$$R(x) \sim \frac{200000 \ln(1 + x)}{\ln(101)}$$

indicating diminishing (logarithmic) returns.

2.2 Scale Function

$$S(x) = 0.0699514x + 0.00345455$$

This is linear in allocation.

2.3 Speed Function (Stochastic Model)

Speed is rank-based:

- Highest allocation $\rightarrow 0.9$
- Lowest allocation $\rightarrow 0.1$

Assume competitors' allocations:

$$p_{\text{other}} \sim \text{Uniform}[0, 100]$$

Then:

$$\Pr(p_{\text{other}} > p) = \frac{100 - p}{100}$$

Expected multiplier:

$$E[V(p)] \approx 0.9 - 0.8 \cdot \Pr(p_{\text{other}} > p)$$

Substitute:

$$E[V(p)] = 0.9 - 0.8 \left(\frac{100 - p}{100} \right)$$

$$E[V(p)] = 0.1 + 0.8 \cdot \frac{p}{100}$$

3 Objective Function

$$\text{PnL} = R(r) \cdot S(s) \cdot \left(0.1 + 0.8 \frac{p}{100}\right) - 500(r + s + p)$$

4 Optimization Approach

We solve:

$$\max_{r,s,p \in Z} \text{PnL}$$

subject to:

$$r + s + p \leq 100$$

Method:

- Full brute-force enumeration
- Integer grid search over all feasible (r, s, p)

5 Optimal Solution

$$r = 16, \quad s = 48, \quad p = 36$$

5.1 Evaluate Components

$$R(16) \approx 122{,}780$$

$$S(48) = 0.0699514(48) + 0.00345455 \approx 3.361$$

$$E[V(36)] = 0.1 + 0.8 \cdot \frac{36}{100} = 0.388$$

5.2 Gross Output

$$\text{Gross} = 122{,}780 \times 3.361 \times 0.388$$

$$\text{Gross} \approx 160{,}119$$

5.3 Cost

$$\text{Cost} = 500(16 + 48 + 36) = 500 \cdot 100 = 50{,}000$$

5.4 Final PnL

$$\text{PnL} = 160{,}119 - 50{,}000 = 110{,}119$$

6 Conclusion

Optimal allocation:

$$\boxed{16\%\text{ Research},\quad 48\%\text{ Scale},\quad 36\%\text{ Speed}}$$

Key insights:

- Research exhibits diminishing returns (logarithmic growth)
- Scale provides linear amplification
- Speed contributes via expected rank advantage
- Full budget utilization is optimal since marginal gains exceed cost